

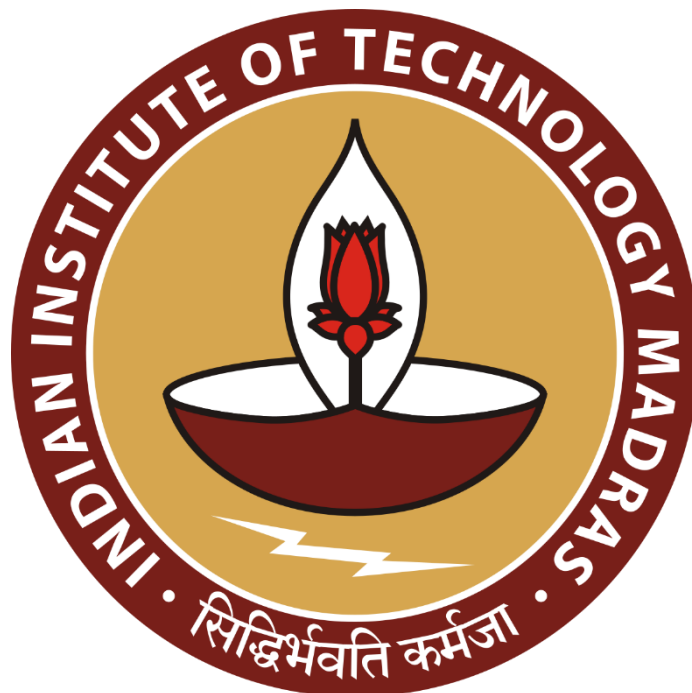
# **Optimizing Inventory Management and Sales Performance in a Retail Pharmacy using Data-Driven Techniques**

A Proposal report for the BDM capstone Project

Submitted by

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## **Declaration Statement**

I am working on a Project titled “Optimizing Inventory Management and Sales Performance in a Retail Pharmacy using Data-Driven Technique”. I extend my appreciation to **Ganesh Medicals**, for providing the necessary resources that enabled me to conduct my project.

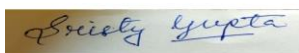
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis has been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



Signature of Candidate: **(Digital Signature)**

Name: Sristy Gupta

Date: 10-11-25

# **Optimizing Inventory Management and Sales Performance in a Retail Pharmacy using Data-Driven Techniques**

## **Executive Summary:**

The project focuses on a small pharmacy shop named Ganesh Medicals located at Bajla Chowk, Deoghar .The business is B2C and deals in the retail sale of medicines, general pharmaceuticals and essentials healthcare products .The pharmacy serves both regular walk-in and prescription customers, maintaining a wide range of medicines.

The pharmacy faces persistent operational challenges, such as maintaining optimal stock levels, adjusting to seasonal demand shifts, and preventing losses from expired products. Additional concerns include irregular purchase patterns mismatch between demand and stock levels, and limited visibility into fast moving and slow-moving items.

The issues will be addressed by analyzing historical sales and inventory data using a wide range of analytical approaches to obtain a fruitful outcome. The approach may include:

- Descriptive analysis to understand overall sales and purchase patterns across the year.
- Trend analysis and seasonality checks to identify sales patterns and seasonal fluctuations.
- Demand forecasting to estimate future requirements

The expected outcomes developed from this analysis will help the pharmacy to reduce capital tied up in excess inventory while ensuring that high-demand products remain consistently available.

## **Organization Background:**

The pharmacy that I am working with is named Ganesh Medicals located at Bajla Chowk, Deoghar , operates as a B2C small retail outlet; which deals in the sale of medicines, general pharmaceuticals and essentials healthcare products .The pharmacy caters to both regular walk-in and prescription customers, maintaining a wide range of medicines.

The shop is managed by Mr. Kedia who oversees daily operations and customer service.

Each month, the pharmacy generates average revenue ranging from ₹90,000 to ₹1,00,000.

Staffing expenses typically range from ₹20,000 to ₹30,000. Operating costs are kept

efficient, with no rent liability and monthly expenses for electricity (₹5,000), printer and stationery supplies, and carry bags (₹4,000) excluding inventory.

Inventory is sourced from a network of regional suppliers and keeps detailed digital records for both inventory and sales, which supports systematic tracking of stock movement. The Management's primary concern is to maintain optimal stock levels and prevent losses, against the backdrop of fluctuating market demand and the strict control of and medicine expiry.

By partnering for this project, the pharmacy seeks to implement data-driven strategies to become more responsive and resilient, thereby ensuring business continuity and improved service quality.

## **Problem Statements:**

### **1.1 Inventory Inefficiency and Expiry-Linked Losses:**

The pharmacy lacks a data-driven inventory planning process, resulting in overstocking, stock outs and medicines approaching expiry faster than they are sold. This mismatch between purchases and actual sales limits working-capital efficiency and reduces profitability.

### **1.2 Seasonality-Driven Demand Misalignment:**

Seasonal fluctuations in customer demand are not reflected in procurement decisions, leading to inconsistent stock availability during peak disease periods and excess inventory during off season months.

### **1.3 Supplier Variability & Procurement Constraints:**

Irregular supplier delivery patterns, varying trade discounts, and purchase decisions that do not incorporate sales trends create planning inefficiencies that impact product availability and cash-flow stability.

## **Background of the Problem**

Frequent fluctuations in customer demand, inconsistent supply schedules, and the inherently perishable nature of pharmaceutical products are at the heart of the operational difficulties faced by the pharmacy. Stock outs of popular medicines damage customer trust, result in lost business, and risk losing regular clientele to competitors. On the other hand, excess

accumulation of slow-moving and seasonal items ties up resources and amplifies financial pressure, particularly when nearing expiry.

Seasonality such as higher sales of certain products during monsoon or flu season—is not systematically reflected in order patterns, further aggravating the inventory mismatch.

Supplier-related uncertainties, delays, and partial deliveries compound these issues, making it difficult for the management to maintain a smooth flow of goods.

Internally, decisions are often based on heuristics or past experiences rather than on predictive analytics. The lack of scientific forecasting and precision in inventory optimization hinders profitability and sustainability. A more structured, data-driven approach is required to address these interconnected challenges.

## **Problem Solving Approach**

This project will follow a structured, step-wise methodology to address the identified challenges:

### **1. Data Collection & Preparation:**

Aggregate and clean primary inventory and sales data. Handle missing values, standardize formats, and identify relevant variables for analysis.

### **2. Descriptive Analytics:**

Develop dashboards and summaries to visualize stock outs, overstock, sales velocity, and expiry events over time. Identify SKUs with chronic availability or wastage issues.

### **3. Seasonal Demand & Trend Analysis:**

Conduct time series analysis and decompositions to uncover seasonal sales trends. Cluster products based on their sales cycles to predict high-demand windows.

### **4. Inventory Forecasting:**

Apply moving average, exponential smoothing, and (where data supports) ARIMA models to forecast future inventory needs, especially for critical SKUs.

### **5. Optimization Modeling:**

Simulate various purchasing strategies, considering supplier lead times, minimum order quantities, and seasonality to minimize out-of-stock and overstock scenarios, thereby optimizing cash flow.

### **6. Expiry and Wastage Management:**

Cross-reference expiry dates with sales projections to highlight products at risk of wastage and develop strategies for proactive promotions or returns.

**7. Supplier Reliability Assessment:**

Analyze historical supplier delivery data to rank vendors on timeliness and consistency, advising on preferred partnerships or alternative sources.

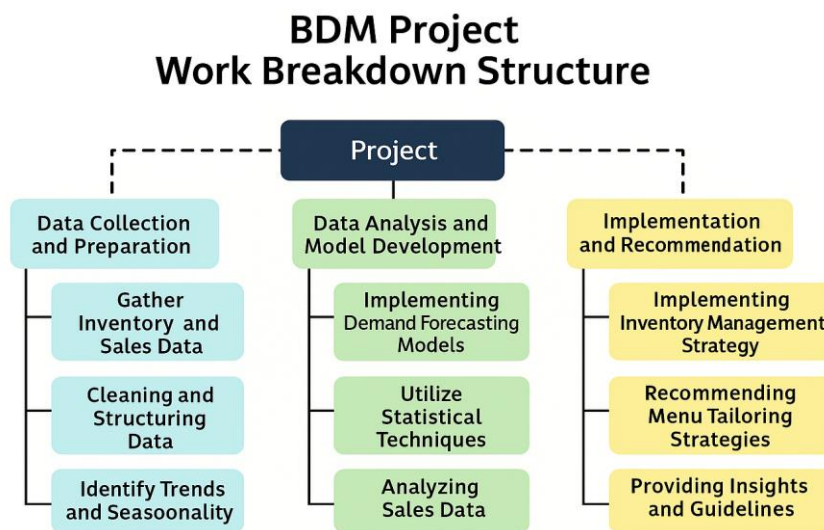
**8. Recommendations & Implementation Plan:**

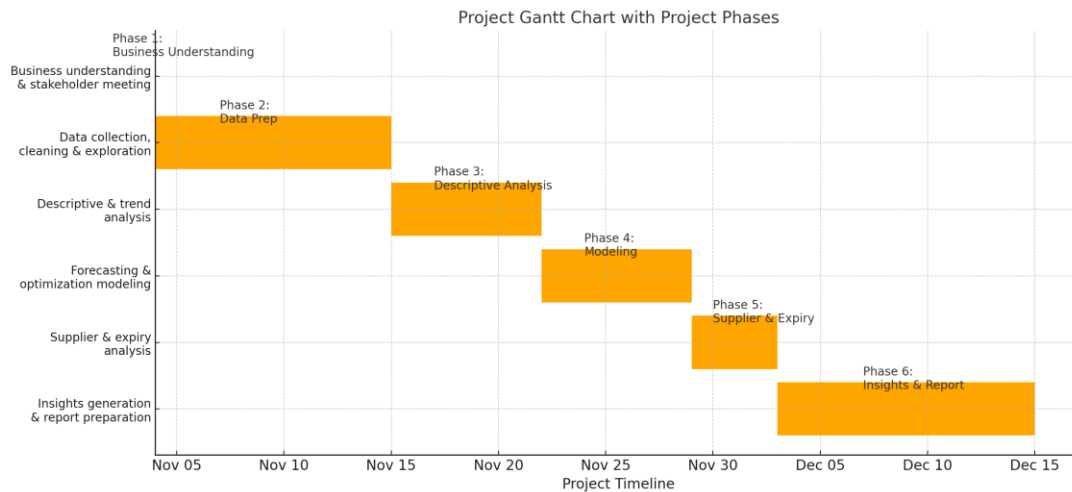
Formulate actionable, data-backed recommendations for inventory policy, procurement scheduling, supplier management, and systematized expiry control.

**Tools:**

Predominantly Excel and Python (with NumPy, Pandas, Matplotlib, and forecasting libraries). Field evidence including process charts, meeting notes, and sample purchase workflows will be documented.

## 1 Expected Timeline





## Expected Outcome:

Successful execution of this project is anticipated to yield:

- Improved ability to predict and fulfill seasonal and ongoing product demand.
- Substantial reduction in the quantity and value of expired or wasted stock.
- Streamlined purchasing strategies that free up working capital without sacrificing service quality.
- Enhanced negotiation and reliability tracking of suppliers, reducing risk from inconsistent deliveries.
- A comprehensive, repeatable analytics framework for ongoing inventory and sales management.

Ultimately, the adoption of these recommendations is expected to elevate both the business's financial performance and the quality of service for its customers.



